



Counting and enumerating independent sets with applications to combinatorial optimization problems

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Abstract

Counting and enumerating maximal and maximum independent sets are well-studied problems in graph theory. In this paper we introduce methods to count and enumerate maximal/maximum independent sets in threshold graphs and k -threshold graphs and improve former results for these problems. The results can be applied to combinatorial optimization problems, and in particular to different variations of the knapsack problem. As feasible solutions for instances of those problems correspond to independent sets in threshold graphs and k -threshold graphs, we obtain polynomial time results for special knapsack and multidimensional knapsack instances. Also, we show lower and upper bounds for the number of necessary bins in several bin packing problems.

Keywords Knapsack problem · Multidimensional knapsack problem · Threshold graphs · Independent sets

Mathematics Subject Classification 05C85 · 05A15 · 05C69

1 Introduction

The knapsack problem is one of the most famous NP-hard tasks in combinatorial optimization. In the knapsack problem (KP) we are given a set A of n items. Every item j has a profit p_j and a size s_j . Further, there is a capacity c for the knapsack. The task is to choose a subset A' of A , such that the total profit of A' is maximized

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while the total size of A' is at most c . The d -dimensional knapsack problem (d -KP) generalizes the knapsack problem by using items with d -dimensional sizes.

Different techniques for solving hard problems were successfully applied to the knapsack problem. Among these are pseudo-polynomial algorithms, approximation algorithms, and integer programming, see Fréville (2004) for a survey. These results lead to several parameterized algorithms, see Gurski et al. (2019). Chvátal and Hammer (1977) considered d -KP instances having zero-one sizes given by some matrix A , such that a corresponding graph model $G(A)$ is a threshold graph.

In this paper we use maximal independent sets in order to solve combinatorial optimization problems. Enumerating and counting maximal independent sets in graphs are frequently studied problems in the field of special graph classes. In general, every graph with n vertices has at most $3^{n/3}$ maximal independent sets (Moon and Moser 1965) and each of these sets can be generated in time in $\mathcal{O}(n^2)$ (Paull and Unger 1959). Thus, all maximal independent sets can be found in time in $\mathcal{O}^*(3^{n/3}) = \mathcal{O}^*(1.4423^n)$. Further, it has been shown in Tsukiyama et al. (1977) that enumerating all maximal independent sets in a graph G with n vertices and m edges can be done in time in $\mathcal{O}(n \cdot m \cdot \text{mis}(G))$, where $\text{mis}(G)$ denotes the number of maximal independent sets in G . Even for special graph classes it is $\#P$ -complete¹ to count the number of maximal independent sets, e.g. regular graphs, planar bipartite graphs of maximum degree 5 (Vadhan 2001), and chordal graphs (Okamoto et al. 2008). Considering the running time with respect to the size of the output, it turns out that enumeration problems are often easier than counting problems. There are special graphs for which enumerating all maximal independent sets can be done in polynomial or even constant time with respect to the size and number of independent sets of the input graph, e.g. interval graphs, circular arc graphs, and chordal graphs (Leung 1984).

We propose new methods to count and enumerate maximal independent sets in different graph classes. In Sect. 2.2 we consider split graphs, since these results can also be applied to threshold graphs and provide ideas for the problem on k -threshold graphs. Our methods for threshold graphs shown in Sect. 2.3 improve the result of Ortiz and Villanueva-Ilufi (2004) and also the result of Tsukiyama et al. (1977) applied to threshold graphs. By combining the maximal independent sets of the k covering threshold graphs, in Sect. 2.4 we describe a method to count and enumerate all maximal independent sets in a k -threshold graph in time in $\mathcal{O}(n^{2k+1})$. Our method implies new bounds on the number of maximal independent sets in k -threshold graphs by using the clique number of k covering threshold graphs. For $k = 2$ we can improve these results by considering split partitions for the two covering threshold graphs and obtain all maximal independent sets in a 2-threshold graph in time in $\mathcal{O}(n^3)$. Our results improve existing solutions for the maximum independent set problem on k -threshold graphs from Caprara et al. (2004), if we can bound the vertex degree of the threshold graphs.

Regarding the application of our results to combinatorial optimization, in Sect. 3.1 we introduce different characterizations for knapsack instances having an equivalent

¹ The class of $\#P$ -complete problems is a class of computationally equivalent counting problems which are at least as difficult as NP-complete problems.

(threshold) graph. Doing so, we obtain an equivalence between independent sets of threshold graphs and feasible solutions in the knapsack problem. This allows us to solve every KP instance with n items which has an equivalent graph in time in $\mathcal{O}(n^2)$. In the subsequent Sect. 3.2, we characterize d-KP instances having an equivalent (k -threshold) graph. This allows us to solve every d-KP instance with n items, having an equivalent graph, in time in $\mathcal{O}(n^{2d+1})$. This result generalizes the above mentioned method by Chvátal and Hammer (1977) for solving d-KP instances, since we consider instances with non-negative real valued sizes and capacities, as well as k -threshold graphs as graph models. Furthermore, in Sect. 3.3, we provide a solution for the bin packing problem using equivalent graphs. Initially, the idea was introduced in Caprara et al. (2004). However, the authors did not characterize what type of bin packing instances can be solved this way. In Sect. 3.4, we show lower and upper bounds for the d -dimensional vector packing and d -dimensional bin packing problem.

In the final section, we survey our results on counting and enumerating special independent sets of threshold and k -threshold graphs (cf. Table 3) and discuss the related problem for cliques.

2 Independent sets in special graph classes

2.1 Notations

We work with finite undirected graphs $G = (V, E)$, where V is a finite set of *vertices* and $E \subseteq \{\{u, v\} \mid u, v \in V_G, u \neq v\}$ is a finite set of *edges*.

A graph $G' = (V', E')$ is a *subgraph* of graph $G = (V, E)$ if $V' \subseteq V$ and $E' \subseteq E$. If every edge of E with both end vertices in V' is in E' , we say that G' is an *induced subgraph* of graph G and we write $G' = G[V']$. For $v \in V$ we define the *neighbourhood* of v by $N(v) = \{u \in V \mid \{v, u\} \in E\}$. The *degree* of a vertex $v \in V$, denoted by $\deg(v)$, is the number of neighbours of vertex v in G , i.e., $\deg(v) = |N(v)|$. Every vertex v with $\deg(v) = 0$ is an *isolated vertex* and every vertex v with $\deg(v) = |V| - 1$ is a *dominating vertex* for G . For $v \in V$ and $S \subseteq V$ we define the neighbourhood of v in S by $N(v, S) = \{u \in S \mid \{u, v\} \in E\}$ and the non-neighbourhood of v in S by $\overline{N(v, S)} = S - (N(v, S) \cup \{v\})$. The *(edge) complement graph* $\overline{G}$ of graph G is defined as $\overline{G} = (V, \{\{u, v\} \mid u, v \in V, u \neq v, \{u, v\} \notin E\})$.

An *independent set* of G is a subset V' of V such that there is no edge in G between two vertices from V' . A *maximum independent set* is an independent set of largest size. A *maximal independent set* is an independent set that is not a proper subset of any other independent set. The family of all independent sets (maximum independent sets and maximal independent sets, respectively) of some graph G is denoted by $I(G)$ ($\text{IM}(G)$ and $\text{MIS}(G)$, respectively). The cardinalities of these families are denoted by $i(G)$ ($\text{im}(G)$ and $\text{mis}(G)$, respectively). A *clique* is a set of pairwise adjacent vertices. The size of a maximum clique in G is denoted by $\omega(G)$, and the size of a maximum independent set in G is denoted by $\alpha(G)$.

2.2 Split graphs

A *split graph* is a graph G whose vertex set V can be partitioned into a clique K and an independent set S (either of which may be empty). For the edges between vertices of K and vertices of S there are no restrictions. For a split graph with a partition of its vertex set into a clique K and an independent set S , we call the pair (K, S) a *split partition*.

Theorem 1 (Hammer and Simeone 1981) *Let G be a split graph with split partition (K, S) . Then, exactly one of the three conditions holds.*

1. $|S| = \alpha(G)$ and $|K| = \omega(G)$. In this case the partition is unique.
2. $|S| = \alpha(G)$ and $|K| = \omega(G) - 1$. In this case there is one vertex $x \in S$, such that $K \cup \{x\}$ is a clique.
3. $|S| = \alpha(G) - 1$ and $|K| = \omega(G)$. In this case there is one vertex $x \in K$, such that $S \cup \{x\}$ is an independent set.

2.2.1 Maximal independent sets in split graphs

The special structure of split graphs allows us to count and enumerate all maximal independent sets as follows. The result can also be applied to threshold graphs and even provides ideas for the problem on k -threshold graphs (cf. Theorem 9).

Theorem 2 *For every split graph G the number of maximal independent sets is $\omega(G)$ or $\omega(G) + 1$. For every split graph G with n vertices, which is given by a split partition, all maximal independent sets can be counted and enumerated in time in $\mathcal{O}(\omega(G) \cdot n)$.*

Proof We determine the family of all maximal independent sets of a split graph G with split partition (K, S) . In order to count all maximal independent sets, we assume that K is a maximum clique, i.e., $|K| = \omega(G)$, which can easily be achieved by Theorem 1. We define K' as the subset of vertices in K which are not adjacent to any vertex in S . Then, the maximal independent sets of G can be obtained as follows.

- If $K' = \emptyset$, set S is a maximal independent set.
- If $K' \neq \emptyset$, every vertex $v \in K'$ leads to a maximal independent set $S \cup \{v\}$.
- Every proper subset S' of S is an independent set but not a maximal one. To enumerate all these proper subsets is very inefficient. However, we know that we can extend these sets by adding one vertex from K . Thus, every vertex $v \in K - K'$ together with its non-neighbours in S leads to a maximal independent set, i.e., every vertex $v \in K - K'$ leads to the maximal independent set $\overline{N}(v, S) \cup \{v\}$.

The union of all sets obtained in both steps yields the family of all maximal independent sets of G . Since we assume that we are given a split partition (K, S) for G , the running time for each of the three steps is in $\mathcal{O}(n)$. The approach's correctness follows from the fact that we have considered all possibilities how to extend S , as well as all possibilities how to extend subsets of S with vertices from K .

In order to justify the number of maximal independent sets, we consider the three steps given above. The first step leads to at most one set and the second and third step together lead to $|K| = \omega(G)$ sets. $\square$

By Hammer and Simeone (1981) and Heggenes and Kratsch (2007) split graphs can be recognized and a split partition can be found in linear time.

Corollary 1 *For every split graph G with n vertices and m edges all maximal independent sets can be counted and enumerated in time in $\mathcal{O}(\omega(G) \cdot n + m)$. After $\mathcal{O}(n + m)$ steps of precomputation all independent sets in G can be enumerated in constant time per output.*

2.3 Threshold graphs

We will use the following characterizations for threshold graphs.

Theorem 3 (Mahadev and Peled 1995) *For every graph $G = (V, E)$ the following statements are equivalent.*

1. *There exist non-negative reals w_v , $v \in V$, and T such that for every $U \subseteq V$ it holds that $\sum_{v \in U} w_v \leq T$ if and only if U is an independent set of G .*
2. *Graph G contains no cycle of four vertices C_4 , no path P_4 with four vertices, and no two disjoint edges $2K_2$ as induced subgraph.*
3. *There exist non-negative reals w_v , $v \in V$, and T such that for every two vertices u, v it holds that $w_u + w_v > T$ if and only if $\{u, v\} \in E$.*
4. *Graph G can be constructed from the one-vertex graph K_1 by repeatedly adding an isolated vertex or a dominating vertex.*
5. *Graph G is a split graph with vertex partition $V = S \cup K$ and the neighbourhoods of the vertices of the independent set S are nested, i.e., for $s_1, s_2 \in S$ it holds that $N(s_1) \subseteq N(s_2)$ or $N(s_2) \subseteq N(s_1)$.*

Condition (1.) of Theorem 3 was used by Chvátal and Hammer to define threshold graphs (Chvátal and Hammer 1973, 1977). By Golumbic (1980, p. 221) one can assume that all w_v and T are non-negative integers. By condition (4.) of Theorem 3 every threshold graph $G = (V, E)$ with $V = \{v_1, \dots, v_n\}$ can be constructed from the one-vertex graph by repeatedly adding an isolated vertex or a dominating vertex. A creation sequence for G is a binary string $t = t_1, \dots, t_n$ of length n such that there is a bijection $v : \{1, \dots, n\} \rightarrow V$ with

- $t_i = 1$ if $v(i)$ is a dominating vertex for the graph $G[\{v(1), \dots, v(i)\}]$ and
- $t_i = 0$ if $v(i)$ is an isolated vertex for the graph $G[\{v(1), \dots, v(i)\}]$.

Without loss of generality we define a single vertex to be a dominating vertex, i.e., $t_1 = 1$. By Mahadev and Peled (1995, Fig. 1.4) a creation sequence can be found in linear time.

Lemma 1 (Mahadev and Peled 1995) *Given some threshold graph G with n vertices and m edges, a creation sequence for G can be found in time in $\mathcal{O}(n + m)$.*

Since the vertex set $\{v(i) \mid i = 1 \vee t_i = 0\}$ leads to a maximum independent set and the vertex set $\{v(i) \mid t_i = 1\}$ leads to a maximum clique, the next result is obvious.

Corollary 2 *Let G be a threshold graph with n vertices and m edges. The size of a maximum clique $\omega(G)$ and the size of a maximum independent set $\alpha(G)$ can be determined in time in $\mathcal{O}(n + m)$.*

2.3.1 Maximal independent sets in threshold graphs

Since threshold graphs are split graphs, we can use the method of Theorem 2 to enumerate all maximal independent sets in a threshold graph. From a given creation sequence $t = t_1, \dots, t_n$ a split partition (K, S) can be found in time in $\mathcal{O}(n)$ as follows. Vertex $v(1)$ can be arbitrarily allocated into K or S . For $i > 1$, if $t_i = 0$ vertex $v(i)$ will be allocated into S , and if $t_i = 1$, vertex $v(i)$ will be allocated into K . Furthermore, if we choose $v(1)$ to be in K , this set leads to a maximum clique, i.e., $|K| = \omega(G)$ and $|S| = \alpha(G) - 1$. If we put $v(1)$ into S , this set leads to a maximum independent set, i.e., $|S| = \alpha(G)$ and $|K| = \omega(G) - 1$.

However, we can exploit the creation sequences in order to obtain simpler methods to enumerate all maximal independent sets in threshold graphs.

Theorem 4 *The number of all maximal independent sets in a threshold graph G equals $\omega(G)$. Let G be a threshold graph G with n vertices given by a creation sequence. Then, all maximal independent sets in G can be counted in time in $\mathcal{O}(n)$ and enumerated in time in $\mathcal{O}(\omega(G) \cdot n)$.*

Proof Let G be a threshold graph and $t = t_1, \dots, t_n$ a creation sequence for G , i.e., $t_1 = 1$ and $t_i \in \{0, 1\}$. Then, for every j , such that $t_j = 1$, the vertex set $\{v(j)\} \cup \{v(i) \mid j < i, t_i = 0\}$ results in a maximal independent set. This allows us to compute $\text{MIS}(G)$ by considering t_i for $i = n$ down to 1. If $t_i = 0$, we add vertex $v(i)$ to a set A and if $t_i = 1$ we extend $\text{MIS}(G)$ by the set $\{A \cup \{v(i)\}\}$.

Since the vertices corresponding to 1 in the creation sequence lead to a maximum clique, our method implies that every threshold graph G has exactly $\omega(G)$ maximal independent sets. Furthermore, the sketched method shows that the stated running times hold. $\square$

For certain threshold graphs G we can bound the value of $\omega(G)$, implying that the graphs only have a small number of maximal independent sets. This holds for threshold graphs which are planar, of bounded vertex degree, or uniformly ℓ -sparse.²

Since a creation sequence can be found in time in $\mathcal{O}(n + m)$ by Lemma 1, we obtain the next result by Theorem 4. Further, the time of computation is bounded by the size of the output, and thus, all independent sets in G can be enumerated in constant time per output.

Corollary 3 *Let G be a threshold graph with n vertices and m edges. All maximal independent sets in G can be counted in time in $\mathcal{O}(n + m)$. All maximal independent sets in G can be enumerated in time in $\mathcal{O}(\omega(G) \cdot n + m)$. After $\mathcal{O}(n + m)$ steps of precomputation all independent sets in G can be enumerated in constant time per output.*

Okamoto et al. (2008) states multiple results with respect to counting and enumeration problems for chordal graphs. Especially, results concerning independent sets are provided. Since threshold graphs are chordal, the results concerning upper bounds

² A graph G with n vertices and m edges is ℓ -sparse if $m \leq \ell \cdot n$. It is uniformly ℓ -sparse if every subgraph of G is ℓ -sparse.

also hold for threshold graphs, see Table 3. While counting all maximal independent sets in chordal graphs is #P-complete (Okamoto et al. 2008), the problem is easy for threshold graphs, as Corollary 3 shows.

2.3.2 Maximum independent sets in threshold graphs

Since every maximum independent set is a maximal independent set, our results given in Sect. 2.3.1 can also be used to list all maximum independent sets in a threshold graph. By the method given in the proof of Theorem 4 and removing non-maximum sets, we obtain a method for listing all maximum independent sets in a threshold graph in time in $\mathcal{O}(\omega(G) \cdot n + m)$. Next, we describe an improved method for this purpose.

Theorem 5 *Let G be a threshold graph G with n vertices given by a creation sequence. Then, all maximum independent sets in G can be counted in time in $\mathcal{O}(n)$ and enumerated in time in $\mathcal{O}(\omega(G) \cdot \alpha(G))$.*

Proof Let G be a threshold graph and $t = t_1, \dots, t_n$ a creation sequence for G . Further, let j be the minimal integer, such that $t_j = 0$. Let A be the set of all vertices $v(i)$, such that $t_i = 0$. Since we assume that $t_1 = 1$, we know that $j \geq 2$. Then, for every $i < j$, vertex $v(i)$ together with all vertices in A leads to a maximum independent set in G . This allows us to compute $\text{IM}(G)$ by considering t_i for $i = 1$ to $j - 1$. In every step we have to extend $\text{IM}(G)$ by $\{\{v(i)\} \cup A\}$.

The number of maximum independent sets in a threshold graph is equal to the value of $j - 1 \leq \omega(G)$, and each set has $\alpha(G)$ elements. $\square$

By Lemma 1 we obtain the following result.

Corollary 4 *Let G be a threshold graph with n vertices and m edges. The number of all maximum independent sets in G is at most $\omega(G)$. All maximum independent sets in G can be counted in time in $\mathcal{O}(n + m)$. All maximum independent sets in G can be enumerated in time in $\mathcal{O}(\omega(G) \cdot n + m)$. After $\mathcal{O}(n + m)$ steps of precomputation all independent sets in G can be enumerated in constant time per output.*

The latter result can also be obtained from the fact that threshold graphs are chordal graphs and in Okamoto et al. (2008) it has been shown for these graphs that all maximum independent sets in a chordal graph with n vertices and m edges can be counted in time $\mathcal{O}(n + m)$ and enumerated in time $\mathcal{O}(1)$ per output. The related problem of finding *one* maximum independent set in a threshold graph was solved in Caprara et al. (2004) in time in $\mathcal{O}(n \log n)$. This problem can also be solved by Corollary 2. A remarkable difference between our solutions and those of Caprara et al. (2004) is that we work with graph representations, while the authors of Caprara et al. (2004) use the coefficients occurring in the knapsack instance. Each of these versions can be transformed into the other one in quadratic time.

2.3.3 Independent sets in threshold graphs

Next, we want to enumerate and count all independent sets in a threshold graph. While the number of maximal and maximum independent sets is bounded by the number of

vertices, the number of independent sets can be in $\Theta(2^n)$, e.g. for edgeless graphs or stars with n vertices.

Theorem 6 *Let G be a threshold graph with n vertices given by a creation sequence. All independent sets in G can be counted in time in $\mathcal{O}(n)$. All independent sets in G can be enumerated in time in $\mathcal{O}(n \cdot 2^{n-1})$.*

Proof Let G be a threshold graph and $t = t_1, \dots, t_n$ a creation sequence for G . Further, let G_i be the subgraph of G induced by $\{v(1), \dots, v(i)\}$. Then, it holds that $i(G_1) = 1$. For $i > 1$ and for $t_i = 1$ we have $i(G_i) = i(G_{i-1}) + 1$ and for $t_i = 0$ we have $i(G_i) = 2 \cdot i(G_{i-1}) + 1$. Thus, we can count all independent sets in G in time in $\mathcal{O}(n)$.

Every vertex $v(i)$ leads to an independent set. Further, every vertex $v(i)$ with $t_i = 0$ can be added to an existing independent set. This allows us to compute $I(G)$ by considering t_i for $i = 1$ to n . If $t_i = 1$, we extend $I(G)$ by $\{\{v(i)\}\}$. If $t_i = 0$, we extend $I(G)$ by $\{\{v(i)\}\} \cup \{I \cup \{v(i)\} \mid I \in I(G)\}$. The time for listing all independent sets in G depends on its size and is bounded by $\sum_{i=1}^n i \cdot \binom{n}{i} = n \cdot 2^{n-1}$, the maximum number of independent sets in an edgeless graph. $\square$

Lemma 1 leads to the following result.

Corollary 5 *Let G be a threshold graph with n vertices and m edges. All independent sets in G can be counted in time in $\mathcal{O}(n+m)$. All independent sets in G can be enumerated in time in $\mathcal{O}(n \cdot 2^{n-1})$. After $\mathcal{O}(n+m)$ steps of precomputation all independent sets in G can be enumerated in constant time per output.*

2.4 k -Threshold graphs

In order to characterize graphs which are equivalent to instances of the d -dimensional knapsack problem, we recall the notation of the threshold dimension from Golumbic (1980). The *threshold dimension* of a graph $G = (V, E)$ with n vertices, $t(G)$ for short, is the minimum number k of linear inequalities of the form

$$a_{i,1}x_1 + \dots + a_{i,n}x_n \leq T_i, \quad (1)$$

such that $X \subseteq V$ is an independent set in G if and only if the characteristic vector $(x_1, \dots, x_n)$ of X satisfies the inequalities of type (1) for $i = 1, \dots, k$. By Golumbic (1980, p. 221) we can assume that all $a_{i,j}$ and T_i are non-negative integers.

Theorem 7 (Chvátal and Hammer 1977) *For every graph $G = (V, E)$ the following statements are equivalent.*

1. G has threshold dimension at most k .
2. There are at most k threshold graphs $G_i = (V, E_i)$ such that $E = E_1 \cup \dots \cup E_k$.

Graphs G with $t(G) = 0$ are exactly the edgeless graphs and graphs G with $t(G) \leq 1$ are exactly threshold graphs. A complete characterization of graphs G with

$t(G) \leq 2$ is unknown. For a given graph G it is NP-complete to determine whether it holds that $t(G) \leq 3$, see Yannakakis (1982).

A graph is a k -threshold graph if and only if its threshold dimension is at most k . For $k = 1$ the set of k -threshold graphs is closed under taking complements. This does not hold for $k \geq 2$, since $t(K_{n,n}) = n \neq 2 = t(\overline{K_{n,n}})$ for $n \geq 3$. Consequently, in general the threshold dimension of a graph G is not equal to the threshold dimension of its complement $\overline{G}$. But, the threshold dimension of a graph equals the threshold intersection dimension of its complement graph. For a graph $G = (V, E)$ the *threshold intersection dimension*, $ti(G)$ for short, is the minimum number k of threshold graphs $G_i = (V, E_i)$, $1 \leq i \leq k$, such that $E = E_1 \cap \dots \cap E_k$, see Rival (1985). We (did not find some assigned notation and) call a graph k -threshold intersection if and only if its threshold intersection dimension is at most k .

2.4.1 Maximal independent sets in k -threshold graphs

Next, we show how to enumerate and count the maximal independent sets in a k -threshold graph G .

Theorem 8 *Let G be a k -threshold graph with n vertices, given by an edge set covering of k threshold graphs $G_i = (V, E_i)$ with m_i edges, $1 \leq i \leq k$. Then, all maximal independent sets of G can be enumerated and counted in time in $\mathcal{O}(\sum_{i=1}^k m_i + n \cdot (\prod_{i=1}^k \omega(G_i))^2) \subseteq \mathcal{O}(n^{2k+1})$.*

Proof Let G be a k -threshold graph with n vertices and m edges. Further, let $G_i = (V, E_i)$, $1 \leq i \leq k$, be a covering of k threshold graphs for G and denote by m_i the number of edges in G_i . In order to compute $\text{MIS}(G)$, we first compute $\text{MIS}(G_i)$ for $1 \leq i \leq k$ by the method described in the proof of Theorem 4. Next, we compute $\mathcal{I} = \{M_1 \cap \dots \cap M_k \mid (M_1, \dots, M_k) \in \text{MIS}(G_1) \times \dots \times \text{MIS}(G_k)\}$. Finally, we remove all sets from $\mathcal{I}$, which are subsets of other sets in $\mathcal{I}$.

Using Corollary 3, the running time to compute $\text{MIS}(G_i)$ for $1 \leq i \leq k$ can be bounded by $\mathcal{O}(\sum_{i=1}^k (\omega(G_i) \cdot n + m_i)) = \mathcal{O}(n \cdot \sum_{i=1}^k \omega(G_i) + \sum_{i=1}^k m_i)$.

Since every family $\text{MIS}(G_i)$ consists of $\omega(G_i) \leq n$ sets, there are at most $\prod_{i=1}^k \omega(G_i) \leq n^k$ tuples $(M_1, \dots, M_k)$ in $\text{MIS}(G_1) \times \dots \times \text{MIS}(G_k)$. For every tuple $(M_1, \dots, M_k)$ the intersection $M_1 \cap \dots \cap M_k$ can be computed as follows. We are given k subsets of a set (the vertex set of G) of n elements. These k subsets are merged into one set M of at most $k \cdot n$ elements. We can sort M using counting sort in time in $\mathcal{O}(k \cdot n + n) = \mathcal{O}(k \cdot n)$. Then, an element of M belongs to the intersection if and only if it occurs k times consecutively in the sorted list M . This can be checked by comparing every first element on position i with the element on position $i + k - 1$. Thus, one intersection $M_1 \cap \dots \cap M_k$ can be computed in time in $\mathcal{O}(k \cdot n)$ and the set of all intersections $\mathcal{I}$ can be computed in time in $\mathcal{O}(k \cdot n \cdot (\prod_{i=1}^k \omega(G_i)))$.

Then, we have to eliminate non-maximal subsets within $\mathcal{I}$, where $|\mathcal{I}| \leq \prod_{i=1}^k \omega(G_i) \leq n^k$. For every pair $(I, J) \in \mathcal{I} \times \mathcal{I}$ we can sort I and J and afterwards verify whether I is a subset of J in time in $\mathcal{O}(n)$. Thus, we can obtain $\text{MIS}(G)$ from $\mathcal{I}$ in time in $\mathcal{O}(n \cdot (\prod_{i=1}^k \omega(G_i))^2)$.

Assuming $m_i \geq 1$, we obtain $\omega(G_i) \geq 2$ for $1 \leq i \leq k$ and it follows that the overall running time is in

$$\begin{aligned} & \mathcal{O} \left(n \cdot \sum_{i=1}^k \omega(G_i) + \sum_{i=1}^k m_i + k \cdot n \cdot \left(\prod_{i=1}^k \omega(G_i) \right) + n \cdot \left(\prod_{i=1}^k \omega(G_i) \right)^2 \right) \\ & \subseteq \mathcal{O} \left(n \cdot \prod_{i=1}^k \omega(G_i) + \sum_{i=1}^k m_i + \left(\prod_{i=1}^k \omega(G_i) \right) \cdot n \cdot \left(\prod_{i=1}^k \omega(G_i) \right) \right. \\ & \quad \left. + n \cdot \left(\prod_{i=1}^k \omega(G_i) \right)^2 \right) \subseteq \mathcal{O} \left(\sum_{i=1}^k m_i + n \cdot \left(\prod_{i=1}^k \omega(G_i) \right)^2 \right) \\ & \subseteq \mathcal{O} \left(n^{2k+1} \right). \end{aligned}$$

The correctness holds as follows. Every independent set S in G is also an independent set in G_i for every $1 \leq i \leq k$ by the definition of G . Thus, every independent set S in G is a subset of some *maximal* independent set M_i in G_i for every $1 \leq i \leq k$. Therefore, every independent set S in G is a subset of the intersection $M_1 \cap \dots \cap M_k$ for some maximal independent set M_i in G_i for every $1 \leq i \leq k$. Since every such intersection $M_1 \cap \dots \cap M_k$ is an independent set in G and we remove the non-maximal independent sets from the set of all these intersections in the last step of our method, we exactly create the set of all maximal independent sets of G . $\square$

The method given in the proof of Theorem 8 implies the following bound.

Corollary 6 *Let G be a k -threshold graph and $G_i = (V, E_i)$, $1 \leq i \leq k$, a covering by k threshold graphs for G . Then, G has at most $\prod_{i=1}^k \omega(G_i)$ maximal independent sets.*

In order to show that the bound $\text{mis}(G) \leq \prod_{i=1}^k \omega(G_i)$ can be achieved, we consider G as the disjoint union of k complete graphs K_ℓ . Then, for $\ell \geq 2$ the graph G has threshold dimension k . Further, G leads to $\ell^k = \prod_{i=1}^k \omega(K_\ell)$ maximal independent sets.

For several graphs, such as planar graphs, graphs of bounded vertex degree, and uniformly ℓ -sparse graphs, we can bound the size of the largest complete subgraph. Thus, in connection with Corollary 6 we obtain bounds for the number of maximal independent sets in special k -threshold graphs.

The method given in the proof of Theorem 8 results in all maximal independent sets for k -threshold graphs and yields an interesting bound for the number of maximal independent sets using the clique number of k suitable covering threshold graphs (Corollary 6). The main drawback of the method given in the proof of Theorem 8 is the last step, in which we have to remove non-maximal subsets. For better solutions to enumerate all maximal independent sets in k -threshold graphs we consider the case $k = 2$. The main idea is to partition the vertex set of a 2-threshold graph into three cliques and one independent set. This allows us to generalize the idea for split graphs given in the proof of Theorem 2.

Table 1 Partition of the vertex set of a 2-threshold graph in the proof of Theorem 9

	K_2	S_2
K_1	K	A
S_1	B	S

Theorem 9 *All maximal independent sets in a 2-threshold graph with n vertices can be enumerated in time in $\mathcal{O}(n^3)$ and counted in time $\mathcal{O}(n^2)$.*

Proof Let $G = (V, E)$ be a 2-threshold graph. Then, there are two threshold graphs $G_1 = (V, E_1)$ and $G_2 = (V, E_2)$ that cover the edge set of G , i.e., $E = E_1 \cup E_2$. Let (K_1, S_1) be a split partition of G_1 and (K_2, S_2) be a split partition of G_2 . Such partitions can be found in linear time by Hammer and Simeone (1981), Heggenes and Kratsch (2007). We consider the intersections $K = K_1 \cap K_2$, $S = S_1 \cap S_2$, $A = K_1 \cap S_2$, and $B = S_1 \cap K_2$, see Table 1. Then, K , A , and B are cliques in G and S is an independent set in G . Since (K_1, S_1) and (K_2, S_2) are partitions of V , it holds that $V = K \cup S \cup A \cup B$. We define K' (A' , B' , respectively) as the subset of vertices in K (A , B , respectively) which are not adjacent to any vertex in S .

The maximal independent sets of G can be found as follows.

1. Set S is an independent set. But S need not be maximal (even when S_1 and S_2 are maximal). That is, we have to verify whether we can add vertices from A' , B' , or K' to S . Since K_1 and K_2 are cliques in G_1 and G_2 , they are also cliques in G . Thus, $A \cup K$ and $B \cup K$ are cliques in G , too. Hence, we cannot add a vertex from A' and a vertex from K' to an independent set. The same holds for B' and K' and for all three sets A' , B' , and K' . We can either add a vertex from K' or at most two nonadjacent vertices from $A' \cup B'$.
 - (a) If $K' = A' = B' = \emptyset$, then S is a maximal independent set.
 - (b) If K' is non-empty, we obtain a maximal independent set $S \cup \{v\}$ for every $v \in K'$.
 - (c) If either A' or B' is empty, the other one will be treated as K' .
 - (d) If A' and B' are non-empty, we obtain a maximal independent set $S \cup \{v_1, v_2\}$ for every $(v_1, v_2) \in A' \times B'$ with $\{v_1, v_2\} \notin E$.
 - (e) If A' and B' are non-empty, we obtain a maximal independent set $S \cup \{v_1\}$ for every $v_1 \in A'$ such that v_1 is adjacent to every vertex in B' .
 - (f) If A' and B' are non-empty, we obtain a maximal independent set $S \cup \{v_2\}$ for every $v_2 \in B'$ such that v_2 is adjacent to every vertex in A' .

In these cases we have to add up to two vertices into S to obtain a maximal independent set.

2. Every proper subset of S is an independent set but not a maximal independent set. However, the possibilities how we can add vertices from $V - S$ to subsets of S are restricted as mentioned in case (1.). Obviously, a vertex $v \in V - S$ can only be added to the (possibly empty) set of its non-neighbours in S , which we denote by $\overline{N(v, S)}$.

- (a) If $K - K'$ is non-empty, we obtain a maximal independent set $\overline{N(S, v)} \cup \{v\}$ for every $v \in K - K'$.
- (b) If either $A - A'$ or $B - B'$ is empty, the other one will be treated as $K - K'$.
- (c) If $A - A'$ and $B - B'$ are non-empty, we obtain a maximal independent set $(\overline{N(v_1, S)} \cap \overline{N(v_2, S)}) \cup \{v_1, v_2\}$ for every $(v_1, v_2) \in ((A - A') \times (B - B')) \cup (A' \times (B - B')) \cup ((A - A') \times B')$ with $\{v_1, v_2\} \notin E$.
- (d) We obtain a maximal independent set $\overline{N(v_1, S)} \cup \{v_1\}$ for every $v_1 \in A - A'$ such that there is no $v_2 \in B - B'$ with $\{v_1, v_2\} \notin E$ and $N(v_2, S) \subseteq N(v_1, S)$ (in order to ensure maximality of the sets).
- (e) We obtain a maximal independent set $\overline{N(v_2, S)} \cup \{v_2\}$ for every $v_2 \in B - B'$ such that there is no $v_1 \in A - A'$ with $\{v_1, v_2\} \notin E$ and $N(v_1, S) \subseteq N(v_2, S)$ (in order to ensure maximality of the sets).

In these cases we have to add up to two vertices into a subset of S to obtain a maximal independent set.

By our construction all independent sets obtained in both steps are maximal. The union of all these sets leads to the family of all maximal independent sets of G . Each of the eleven conditions can be verified in time in $\mathcal{O}(n^2)$ and leads to $\mathcal{O}(n^2)$ maximal independent sets of size $\mathcal{O}(n)$. The proof's correctness follows from the fact that we considered all possibilities how to extend the independent set S as well as all subsets of S with vertices of K , A , and B . $\square$

The partition of the vertex set of a 2-threshold graph in the proof of Theorem 9 motivates to look at the following generalization of split graphs introduced in Brandstädt (1996). A graph is a (k, ℓ) -graph if its vertex set can be partitioned into k independent sets and ℓ cliques. There is no restriction for the edges between vertices of these sets. Thus, $(2, 0)$ -graphs are exactly bipartite graphs, $(k, 0)$ -graphs are exactly k -colorable graphs, and $(1, 1)$ -graphs are exactly split graphs. The partition given in the proof of Theorem 9 shows that 2-threshold graphs are special $(1, 3)$ -graphs and, more generally, k -threshold graphs are special $(1, 2^{k-1})$ -graphs. Furthermore, k -threshold intersection graphs are special $(2^{k-1}, 1)$ -graphs.

2.4.2 Maximum independent sets in k -threshold graphs

Following the ideas of Sect. 2.3.2, our results can also be used to list all maximum independent sets in k -threshold graphs. Omitting the last step of the method given in the proof of Theorem 8 and removing non-maximum sets, we obtain a method with running time $\mathcal{O}(\sum_{i=1}^k m_i + k \cdot n \cdot (\prod_{i=1}^k \omega(G_i))) \subseteq \mathcal{O}(k \cdot n^{k+1})$ for listing all maximum independent sets in a k -threshold graph.

Corollary 7 *All maximum independent sets in a k -threshold graph G with n vertices, given by an edge set covering of k threshold graphs $G_i = (V, E_i)$ on m_i edges, $1 \leq i \leq k$, can be enumerated and counted in time in $\mathcal{O}(\sum_{i=1}^k m_i + k \cdot n \cdot (\prod_{i=1}^k \omega(G_i))) \subseteq \mathcal{O}(k \cdot n^{k+1})$. The size of a maximum independent set, i.e., $\alpha(G)$, in a k -threshold graph G can be computed in the same time.*

The related problem of finding *one* independent set of maximum size in a k -threshold graph was solved in Caprara et al. (2004) in time in $\mathcal{O}(n \log n + n^{k-1})$.

Especially when we can bound the vertex degree of the threshold graphs (cf. Sect. 2.4.1) our results are much better than the previously mentioned ones.

2.4.3 Independent sets in k -threshold graphs

Remark 1 Counting and enumerating independent sets in k -threshold graphs G can be done using the independent sets of k covering threshold graphs for G , which can be found with the poof of Theorem 6. Since this leads to a running time in $\mathcal{O}(k \cdot n \cdot 2^{n-k})$, it would be more efficient to check all 2^n possible independent sets one after the other.

3 Applications to combinatorial optimization problems

Given some combinatorial optimization problem Π and an instance I of Π we denote by $OPT(\Pi, I)$, $OPT(I)$ for short, the value of an optimal solution for Π on input I . The results of the previous section can be used to compute the value of $OPT(I)$ or bounds for $OPT(I)$ for several problems Π in polynomial time.

Next, we apply our results for finding all maximal independent sets in threshold graphs and k -threshold graphs on knapsack problems and bin packing problems. Our solutions for the knapsack problems are widely independent from the profits, since we compute all maximal feasible solutions and only compare their profits in a final step.

3.1 Knapsack problem

Name: MAX KNAPSACK (MAX KP)

Instance: A set $A = \{a_1, \dots, a_n\}$ of n items, for every item a_j there is a size s_j and a profit p_j . Further, there is a capacity c for the knapsack.

Task: Find a subset $A' \subseteq A$ such that the total profit of A' is maximized and the total size of A' is at most c .

The parameter n is assumed to be a positive integer and the parameters p_j , s_j , and c are assumed to be non-negative reals. Let I be an instance for MAX KP. Every subset A' of A such that $\sum_{a_j \in A'} s_j \leq c$ is a *feasible solution* of I . A feasible solution which is not a subset of another feasible solution is called *maximal*. An instance I for MAX KP and a graph $G = (V, E)$ are *equivalent*, if there is a bijection $f : A \rightarrow V$ such that $A' \subseteq A$ is a feasible solution of I if and only if $f(A') := \{f(a_j) \mid a_j \in A'\}$ is an independent set of G .

Condition (1.) of Theorem 3 implies a characterization for graphs which are equivalent to instances of MAX KP.

Observation 1 A graph is a threshold graph if and only if it has an equivalent instance for MAX KP.

3.1.1 Knapsack problems and threshold graphs

Lemma 2 For a given threshold graph with n vertices an equivalent instance for MAX KP can be constructed in time in $\mathcal{O}(n^2)$.

Proof Let G be a threshold graph with n vertices and m edges. By Lemma 1 we can find a creation sequence for G in time in $\mathcal{O}(n + m)$ and by the algorithm given in part (4) $\Rightarrow$ (1) of the proof of Theorem 1.2.4 in Mahadev and Peled (1995) we obtain an instance I for MAX KP in time in $\mathcal{O}(n^2)$. $\square$

Defining an equivalent graph from an instance for MAX KP is more difficult, since not every MAX KP instance possesses us to give an equivalent graph. Let I be an instance for MAX KP on item set A . If A has a subset A' of at least three items such that for every two different items a_i and a_j of A' it holds that $s_i + s_j \leq c$ and $\sum_{a_i \in A'} s_i > c$, then the corresponding instance does not have an equivalent graph. Avoiding this situation will now characterize MAX KP instances having an equivalent graph. Let $\mathcal{I}$ be the set of all instances of MAX KP. We define a Boolean property $P : \mathcal{I} \rightarrow \{\text{true}, \text{false}\}$ for an instance $I \in \mathcal{I}$ by

$$P(I) = \bigvee_{A' \subseteq A} \left(\left(\bigvee_{a_j, a_{j'} \in A'} s_j + s_{j'} \leq c \right) \Rightarrow \sum_{a_j \in A'} s_j \leq c \right).$$

The idea of this property is to ensure feasibility in the case of independence within A . Although independence has to hold within a graph, this will be useful. For subsets A' with $|A'| \leq 2$ the property is always true. Since we assume that $j \neq j'$, the implication from the right to the left is always true.³ Furthermore, for an instance I of MAX KP we define the graph $G(I) = (V(I), E(I))$ by

$$V(I) = \{v_j \mid a_j \in A\} \quad \text{and} \quad E(I) = \{\{v_j, v_{j'}\} \mid s_j + s_{j'} > c\}.$$

In general, $G(I)$ is not equivalent to I since not every instance of MAX KP has an equivalent graph. But, by Condition (3.) of Theorem 3 and by our assumptions for instances of MAX KP the graph $G(I)$ is even a threshold graph.

Next, we prove characterizations for instances I of MAX KP having an equivalent graph.

Theorem 10 *For every instance I of MAX KP the following conditions are equivalent.*

1. *Instance I has an equivalent graph.*
2. *Instance I satisfies property $P(I)$.*
3. *Instance I is equivalent to graph $G(I)$.*
4. *Instance I has an equivalent threshold graph.*

Proof (1) $\Rightarrow$ (2): Let I be some instance for MAX KP. In order to give a proof by contradiction, we assume that I does not satisfy property $P(I)$. That is, there is some $A' \subseteq A$ such that $\forall_{a_j, a_{j'} \in A'} s_j + s_{j'} \leq c$ and $\sum_{a_j \in A'} s_j > c$. Then, there is some set $A'' \subseteq A'$ and $a_\ell \in A''$ such that $\sum_{a_j \in A''} s_j > c$ and $\sum_{a_j \in A'' - \{a_\ell\}} s_j \leq c$. That means that for every $a_j \in A''$ the set $\{a_j, a_\ell\}$ is a feasible solution for I and $A'' - \{a_\ell\}$ is also a feasible solution for I , but A'' is not a feasible solution for I .

³ For $A = A' = \{a_1, a_2\}$, $s_1 = 2$, $s_2 = 7$, $c = 10$, the right condition is true, but for $j = j' = 2$ the left condition is not true.

Every bijection $f : A \rightarrow V$ certifying an equivalent graph $G = (V, E)$ for I has to map for every $a_j \in A''$ the item set $\{a_j, a_\ell\}$ onto an independent set $\{f(a_j), f(a_\ell)\}$ and item set $A'' - \{a_\ell\}$ onto an independent set $f(A'' - \{a_\ell\}) = f(A'') - f(\{a_\ell\})$. But then $f(A'')$ is an independent set in G , while A'' is not a feasible solution for I .

(2) $\Rightarrow$ (3): Let I be an instance for MAX KP, such that I satisfies property $P(I)$. Let $A' \subseteq A$ be a feasible solution for I , i.e., $\sum_{a_j \in A'} s_j \leq c$. Then, obviously also $\forall_{a_j, a_{j'} \in A'} s_j + s_{j'} \leq c$ holds and by definition of $G(I)$ the vertex set $V' = \{v_j \mid a_j \in A'\}$ is an independent set in $G(I)$. For the reverse direction let $V' \subseteq V$ be an independent set in $G(I)$. Then, for $A' = \{a_j \mid v_j \in V'\}$ it holds that $\forall_{a_j, a_{j'} \in A'} s_j + s_{j'} \leq c$ and since property $P(I)$ is satisfied, this implies that $\sum_{a_j \in A'} s_j \leq c$, i.e., A' is a feasible solution for I . That is, the bijection defined by the definition of $V(I) = \{v_j \mid a_j \in A\}$ verifies that I is equivalent to $G(I)$.

(3) $\Rightarrow$ (4): $G(I)$ is a threshold graph by Condition (3.) of Theorem 3.

(4) $\Rightarrow$ (1): Obvious. □

Corollary 8 1. Let I be an instance for MAX KP which has two equivalent graphs G_1 and G_2 . Then, G_1 is isomorphic to G_2 .
2. Let I be an instance for MAX KP which has an equivalent graph G . Then, G is isomorphic to $G(I)$.

Proof 1. Let $G_i = (V_i, E_i)$ for $i = 1, 2$ be an equivalent graph for I and $f_i : A \rightarrow V_i$ the corresponding bijections. Then, by

$$\begin{aligned} \{v_j, v_{j'}\} \in E_1 &\Leftrightarrow \{f_1^{-1}(v_j), f_1^{-1}(v_{j'})\} \text{ is not a feasible solution for } I \\ &\Leftrightarrow \{f_2(f_1^{-1}(v_j)), f_2(f_1^{-1}(v_{j'}))\} \in E_2 \end{aligned}$$

the isomorphism of G_1 and G_2 follows.

2. Whenever some instance I for MAX KP has an equivalent graph, by Theorem 10 it follows that $G(I)$ is an equivalent graph for I , too. Thus, we obtain the result by (1.). □

3.1.2 Solutions for the knapsack problem

Next, we use our results to solve every instance of MAX KP with n items which has an equivalent graph in time in $\mathcal{O}(n^2)$.

Theorem 11 Let I be an instance for MAX KP with n items which has an equivalent graph. Then, I can be solved in time in $\mathcal{O}(n^2)$.

Proof Let I be some instance for MAX KP with n items which has an equivalent graph. By Theorem 10 instance I is equivalent to graph $G(I)$, which is a threshold graph and can be constructed from I in time in $\mathcal{O}(n^2)$. Thus, the $\omega(G(I)) \leq n$ maximal independent sets in $G(I)$ can be found in $\mathcal{O}(n^2)$ by Corollary 3 and correspond to the maximal feasible solutions of I . For all these solutions we can compute their profits in time in $\mathcal{O}(n)$. □

Knapsack problems have also been studied in connection with conflict graphs (Pferschy and Schauer 2009, 2017; Gurski and Rehs 2019a, b). The solution of knapsack problems with conflict graphs (KPC) are independent sets in the conflict graphs. Since maximal independent sets in the conflict graph do not necessarily correspond to feasible solutions in the knapsack instance, our methods do not imply solutions for knapsack problems with conflict graphs.

3.2 Multidimensional knapsack problem

Name: MAX D-DIMENSIONAL KNAPSACK (MAX D-KP)

Instance: A set $A = \{a_1, \dots, a_n\}$ of n items and a number d of dimensions. Every item a_j has a profit p_j and for dimension i the size $s_{i,j}$. Further, for every dimension i there is a capacity c_i .

Task: Find a subset $A' \subseteq A$ such that the total profit of A' is maximized and for every dimension i the total size of A' is at most the capacity c_i .

In the case of $d = 1$ the MAX D-KP problem corresponds to the MAX KP problem considered in Sect. 3.1. The parameters n and d are assumed to be positive integers and p_j , $s_{i,j}$, and c_i are assumed to be non-negative reals. Let I be an instance for MAX D-KP. Every subset A' of A such that $\sum_{a_j \in A'} s_{i,j} \leq c_i$ for every $i \in [d]$ is a *feasible solution* of I . A feasible solution which is not a subset of another feasible solution is called *maximal*. An instance I for MAX D-KP and a graph $G = (V, E)$ are *equivalent*, if there is a bijection $f : A \rightarrow V$ such that $A' \subseteq A$ is a feasible solution of I if and only if $f(A') := \{f(a_j) \mid a_j \in A'\}$ is an independent set of G .

3.2.1 Multidimensional knapsack problems and k -threshold graphs

Let $G = (V, E)$ be a k -threshold graph. Then, we can define an equivalent instance for MAX D-KP as follows.⁴

Lemma 3 *Let $G = (V, E)$ be a k -threshold graph with n vertices, given by k threshold graphs $G_j = (V, E_j)$, $1 \leq j \leq k$, which cover the edges of G . Then, a MAX D-KP instance I equivalent to G can be constructed in time in $\mathcal{O}(d \cdot n^2)$.*

Proof We assume that we are given k threshold graphs $G_j = (V, E_j)$, $1 \leq j \leq k$, which cover the edge set of G . For each of these graphs G_j we can find a creation sequence in $\mathcal{O}(n^2)$. Then, the method given in part (4) $\Rightarrow$ (1) of the proof of Theorem 1.2.4 in Mahadev and Peled (1995) leads to sizes $s_{j,i}$ and a capacity c_j to obtain an instance of MAX KP. The union of all these k instances (inequalities) results in an equivalent instance for MAX D-KP. $\square$

Next, we consider the problem of defining a graph from a MAX D-KP instance. First, we look at the relation between the existence of an equivalent graph for a MAX D-KP instance I and the existence of an equivalent graph for the d corresponding

⁴ Because of the hardness result in Yannakakis (1982), we assume that we are given k threshold graphs which cover the edge set of G .

Table 2 Examples of MAX 2-KP instances

	$s_{1,1}$	$s_{1,2}$	$s_{1,3}$	$s_{1,4}$	$s_{1,5}$	$s_{1,6}$	c_1	$s_{2,1}$	$s_{2,2}$	$s_{2,3}$	$s_{2,4}$	$s_{2,5}$	$s_{2,6}$	c_2
I	1	2	3	5	5	5	5	5	5	5	5	2	3	5
I'	10	11	12	8	9	10	26	1	1	4	4	8	11	11
I''	1	2	3	5	5	5	5	5	5	5	1	2	3	5
I'''	1	2	3	5	5	5	5	1	2	3	1	2	3	5

MAX KP instances, which are defined as follows. Let I be some MAX D-KP instance with item set $A = \{a_1, \dots, a_n\}$, profits $p_1, \dots, p_n$, sizes $s_{1,1}, \dots, s_{d,n}$, and capacities $c_1, \dots, c_d$. Then, for $i \in [d]$ we denote by I_i the MAX KP instance with item set $A = \{a_1, \dots, a_n\}$, profits $p_1, \dots, p_n$, sizes $s_{i,1}, \dots, s_{i,n}$, and capacity c_i .

Lemma 4 *Let I be an instance of MAX D-KP. If for every dimension $i \in [d]$ the corresponding MAX KP instance I_i possesses an equivalent graph $G_i = (V, E_i)$, the graph $G = (V, E)$ with $E = E_1 \cup \dots \cup E_d$ leads to an equivalent graph for I .*

Proof Let I be some instance of MAX D-KP with item set $A = \{a_1, \dots, a_n\}$, profits $p_1, \dots, p_n$, sizes $s_{1,1}, \dots, s_{d,n}$, and capacities $c_1, \dots, c_d$. By $I_1, \dots, I_d$ we denote the corresponding MAX KP instances. Assume that for every dimension $i \in [d]$ the MAX KP instance I_i possesses an equivalent graph G_i . Every G_i is a threshold graph by Observation 1. Since all graphs are defined on the item set of I , they can be defined on the same vertex set, i.e., $G_i = (V, E_i)$. Then, we obtain a d -threshold graph $G = (V, E)$ by $E = E_1 \cup \dots \cup E_d$. Consequently, $A' \subseteq A$ is a feasible solution for I if and only if A' is a feasible solution for every I_i if and only if $f(A')$ is an independent set of every G_i if and only if $f(A')$ is an independent set of G . Thus, G is equivalent to I . $\square$

On the other hand, it is not possible to carry over the existence of an equivalent graph from I to the instances $I_1, \dots, I_d$. This can be verified by instance I in Table 2, where I_1 has no equivalent graph, I_2 has an equivalent graph, and I has an equivalent graph. Thus, if the instance I_i for some dimension i does not have an equivalent graph, this does not imply that I does not have an equivalent graph. However, if the instance I_i for some dimension i does not have an equivalent graph, it still can follow that I has no equivalent graph, too. This can be verified by instance I' in Table 2, where I'_1 has no equivalent graph, I'_2 has an equivalent graph, and I' has no equivalent graph.

In fact, even if all of the instances $I_1, \dots, I_d$ have no equivalent graph, it is still possible for the original instance I to possess an equivalent graph or not. The first case occurs for instance I'' in Table 2. Both instances I''_1 and I''_2 do not possess an equivalent graph, but instance I'' has an equivalent graph. The second case applies to instance I''' in Table 2. Instances I'''_1 and I'''_2 do not have an equivalent graph and instance I''' does not possess an equivalent graph, too.

Next, we want to characterize MAX D-KP instances having an equivalent graph. Let $\mathcal{I}_d$ be the set of all instances of MAX D-KP. We define a Boolean property

$P : \mathcal{I}_d \rightarrow \{\text{true}, \text{false}\}$ for some instance $I \in \mathcal{I}_d$ by

$$P_d(I) = \bigvee_{A' \subseteq A} \left(\left(\bigvee_{i \in [d]} \bigvee_{a_j, a_{j'} \in A'} s_{i,j} + s_{i,j'} \leq c_i \right) \Rightarrow \left(\bigvee_{i \in [d]} \sum_{a_j \in A'} s_{i,j} \leq c_i \right) \right).$$

The idea of this property is to ensure feasibility in the case of independence within A . Although independence has to hold within a graph, this will be useful. For subsets A' with $|A'| \leq 2$ the property is always true. Since we assume that $j \neq j'$, the implication from the right to the left is always true. Comparing property P_d with property P defined in Sect. 3.1.1, we conclude that if property $P(I_i)$ holds for every $i \in [d]$, then property $P_d(I)$ holds as well. But, the reverse direction does not hold, as examples I'' and I''' in Table 2 prove. Further, for some instance I of MAX D-KP we define the graph $G_d(I) = (V(I), E(I))$ by

$$\begin{aligned} V(I) &= \{v_j \mid a_j \in A\} \quad \text{and} \\ E(I) &= \{\{v_j, v_{j'}\} \mid s_{1,j} + s_{1,j'} > c_1 \vee \dots \vee s_{d,j} + s_{d,j'} > c_d\}. \end{aligned}$$

In general $G_d(I)$ is not equivalent to I , since not every instance of MAX D-KP has an equivalent graph. The graph $G_d(I)$ can also be obtained from the d threshold graphs $G(I_i) = (V(I_i), E(I_i))$, $1 \leq i \leq d$, by $E(I) = E(I_1) \cup \dots \cup E(I_d)$. Although graph $G(I_i)$ does not have to be equivalent to I_i , $1 \leq i \leq k$, we know by Theorem 7 that $G_d(I)$ is a d -threshold graph.

Next, we prove characterizations for instances of MAX D-KP having an equivalent graph.

Theorem 12 *For every instance I of MAX D-KP the following conditions are equivalent.*

1. *Instance I has an equivalent graph.*
2. *Instance I satisfies property $P_d(I)$.*
3. *Instance I is equivalent to graph $G_d(I)$.*
4. *Instance I has an equivalent d -threshold graph.*

Proof (1) $\Rightarrow$ (2): Let I be a MAX D-KP instance. In order to give a proof by contradiction, we assume that I does not satisfy property $P_d(I)$. Consequently, there is some $A' \subseteq A$ such that for every $i \in [d]$ and $\forall_{a_j, a_{j'} \in A'} s_{i,j} + s_{i,j'} \leq c_i$ and there is some dimension $i' \in [d]$ such that $\sum_{a_j \in A'} s_{i',j} > c_{i'}$. Then, there is some set $A'' \subseteq A'$, $a_\ell \in A''$ and $i'' \in [d]$ such that $\sum_{a_j \in A''} s_{i'',j} > c_{i''}$ and for all $i \in [d]$ it holds that $\sum_{a_j \in A'' - \{a_\ell\}} s_{i,j} \leq c_i$. This means that for every $a_j \in A''$ the set $\{a_j, a_\ell\}$ is a feasible solution for I and $A'' - \{a_\ell\}$ is also a feasible solution for I , but A'' is not a feasible solution for I .

Every bijection $f : A \rightarrow V$ certifying an equivalent graph $G = (V, E)$ for I has to map for every $a_j \in A''$ item set $\{a_j, a_\ell\}$ onto an independent set $\{f(a_j), f(a_\ell)\}$ and item set $A'' - \{a_\ell\}$ onto an independent set $f(A'' - \{a_\ell\}) = f(A'') - f(\{a_\ell\})$. But then also $f(A'')$ is an independent set in G while A'' is not a feasible solution for I .

(2) $\Rightarrow$ (3): Let I be an instance for MAX D-KP such that I satisfies property $P_d(I)$. Let $A' \subseteq A$ be a feasible solution for I , i.e., for every $i \in [d]$ it holds that $\sum_{a_j \in A'} s_{i,j} \leq c_i$. Then, obviously for every $i \in [d]$ it holds that $\forall_{a_j, a_{j'} \in A'} s_{i,j} + s_{i,j'} \leq c_i$ and by definition of the edge set $E(I)$ the vertex set $V' = \{v_j \mid a_j \in A'\}$ is an independent set in $G_d(I)$. For the reverse direction let $V' \subseteq V$ be an independent set in $G_d(I)$. Then, for $A' = \{a_j \mid v_j \in V'\}$ and every $i \in [d]$ it holds that $\forall_{a_j, a_{j'} \in A'} s_{i,j} + s_{i,j'} \leq c_i$ and, since property $P_d(I)$ is satisfied, this implies that for every $i \in [d]$ it holds that $\sum_{a_j \in A'} s_{i,j} \leq c_i$, i.e., A' is a feasible solution for I . That is, the bijection defined by the definition of $V(I) = \{v_j \mid a_j \in A\}$ verifies that I is equivalent to $G_d(I)$.

(3) $\Rightarrow$ (4): $G_d(I)$ is a d -threshold graph.

(4) $\Rightarrow$ (1): Obvious. $\square$

The proof of the following result is similar to that of Corollary 8.

Corollary 9 1. Let I be a MAX D-KP instance which has two equivalent graphs G_1 and G_2 . Then, G_1 is isomorphic to G_2 .
 2. Let I be a MAX D-KP instance which has an equivalent graph G . Then, G is isomorphic to $G_d(I)$.

3.2.2 Solutions for the multidimensional knapsack problem

We have shown how to solve all MAX KP instances with n items having an equivalent graph in time in $\mathcal{O}(n^2)$ (cf. Theorem 11) using the method given in the proof of Theorem 4. Next, we want to show a similar result for instances of MAX D-KP using our method for enumerating maximal independent sets in a d -threshold graph.

Theorem 13 Let I be an instance of MAX D-KP with n items which has an equivalent graph. Then I can be solved in time in $\mathcal{O}(n^{2d+1})$.

Proof Let I be an instance for MAX D-KP with n items such that I has an equivalent graph. By Theorem 12 instance I is equivalent to graph $G_d(I)$, which is a d -threshold graph. In order to apply Theorem 8 on $G_d(I)$, we need d covering threshold graphs G_i , $1 \leq i \leq d$, for $G_d(I)$. Such graphs can be obtained by $G_i = G(I_i)$, i.e., the graphs defined in Sect. 3.1.1 for every of the d instances I_i of MAX KP defined in Sect. 3.2.1. Every graph G_i is a threshold graph and can be constructed in time in $\mathcal{O}(n^2)$.

The $\prod_{i=1}^d \omega(G(I_i)) \leq n^d$ maximal independent sets in G can be found in time in $\mathcal{O}(n^{2d+1})$ by Theorem 8 and correspond to the maximal feasible solutions of I . For every of these solutions we can compute its profit in time in $\mathcal{O}(n)$. $\square$

A related approach to solve special MAX D-KP instances was suggested by Chvátal and Hammer (1977). They consider $m \times n$ zero-one matrices A for which there is a single inequality $\sum_{j=1}^n a_j x_j \leq b$ whose zero-one solutions are exactly the zero-one solutions of the m inequalities $\sum_{j=1}^n a_{i,j} x_j \leq 1$, $i = 1, \dots, m$. For such a matrix A Chvátal and Hammer define a graph $G(A)$ by representing the columns of A as vertices and two vertices are adjacent if and only if the dot product of the corresponding vectors is positive. If $G(A)$ is a threshold graph, the MAX D-KP instance I_A using zero-one sizes with respect to A , capacities $c_j = 1$ and $d = m$ dimensions was solved in

Chvátal and Hammer (1977) within time in $\mathcal{O}(m \cdot n^2)$ by using the split partition of graph $G(A)$.

The sketched results of Chvátal and Hammer (1977) motivate us to consider MAX D-KP instances I which have an equivalent threshold graph G . By Corollary 9 we know that G is isomorphic to $G_d(I)$. Since G is a threshold graph, $G_d(I)$ is also a threshold graph. Thus, instead of the method shown in the proof of Theorem 8, we now can apply the method given in the proof of Theorem 4 on graph $G_d(I)$ in order to list all maximal feasible solutions for I . Since $G_d(I)$ can be defined from I in time in $\mathcal{O}(d \cdot n^2)$, we have shown the following result.

Corollary 10 *Let I be an instance for MAX D-KP with n items, such that I has an equivalent threshold graph. Then, I can be solved in time in $\mathcal{O}(d \cdot n^2)$.*

Comparing the running time $\mathcal{O}(d \cdot n^2)$ with the result in Chvátal and Hammer (1977), we obtain the same running time, but can even handle instances using positive integer valued sizes and capacities. Further, $\mathcal{O}(d \cdot n^2)$ is no longer exponential within d as in Theorem 13, but can only solve more restricted MAX D-KP instances.

Theorem 9 allows to improve Theorem 13 for $d = 2$ dimensions:

Theorem 14 *Let I be a MAX 2-KP instance with n items which has an equivalent graph. Then, I can be solved in time in $\mathcal{O}(n^3)$.*

3.3 Bin packing

In Caprara et al. (2004) threshold graphs were briefly considered in order to give lower bounds for solutions to bin packing problems.⁵ But it is not discussed whether these bin packing instances possess equivalent graphs.

Name: MIN BIN PACKING (MIN BP)

Instance: A set $A = \{a_1, \dots, a_n\}$ of n items and a number n of bins. For every item a_j there is a positive rational size of s_j .

Task: Find n disjoint (possibly empty) subsets $A_1, \dots, A_n$ of A such that the number of non-empty subsets is minimized and the sum of the sizes of the items in each subset is at most 1.

For some instance I of MIN BP two items a_j and $a_{j'}$ can be put into the same subset (bin) if and only if

$$s_j + s_{j'} \leq 1. \quad (2)$$

This motivates to model the compatibleness of the items in I by threshold graphs. Therefore, we consider the inequality

$$s_1 x_1 + s_2 x_2 + \dots + s_n x_n \leq 1. \quad (3)$$

⁵ Please note the different but equivalent definition of edges within threshold graphs in Caprara et al. (2004). By using values w_i for the vertices v_i such that $0 < w_i \leq 1$, the authors in Caprara et al. (2004) define edges $\{v_i, v_j\}$ whenever $w_i + w_j \leq 1$. For our notation see the statement (3.) of Theorem 3.

An instance I for MIN BP and a graph $G = (V, E)$ are *equivalent*, if there is a bijection $f : A \rightarrow V$ such that for every $A' \subseteq A$ the characteristic vector of A' satisfies inequality (3) if and only if $f(A') := \{f(a_j) \mid a_j \in A'\}$ is an independent set of G .

If we assume that instance I has an equivalent graph $G = (V, E)$, then we know by Theorem 3 (1.) that G is a threshold graph. Further, by (2) and (3) two items a_j and $a_{j'}$ can be put into the same bin if and only if $\{v_j, v_{j'}\} \notin E$. Thus, any two items corresponding to the vertices of a clique in G cannot be placed into the same bin. This observation yields $\omega(G) \leq OPT(I)$ as a lower bound for an optimal solution.

Since there is a partition of V into $\chi(G)$ non-empty independent sets V_i , the corresponding item sets $A_i = \{f^{-1}(v) \mid v \in V_i\}$ lead to a solution of I using $\chi(G)$ non-empty subsets A_i . Thus, we know $\omega(G) \leq OPT(I) \leq \chi(G)$. Since every threshold graph is perfect, we conclude that $\omega(G) = OPT(I) = \chi(G)$.

In order to obtain an equivalent graph for a MIN BP instance I , we can proceed as in Sect. 3.1.1 and use $G(I)$ defined by the values of inequality (3).

Theorem 15 *Let I be an instance for MIN BP which has an equivalent graph G . Then, $OPT(I) = \omega(G) = \chi(G) = \omega(G(I)) = \chi(G(I))$.*

3.4 d -Dimensional vector packing and d -dimensional bin packing

Next, we consider the generalization of MIN BP where every item a_j corresponds to a d -dimensional vector $s_j := (s_{1,j}, \dots, s_{d,j})$ and every bin i to the unit d -dimensional vector $(1, \dots, 1)$. A set of items A' fits into a bin if each component of the vector $\sum_{a_j \in A'} s_j$ does not exceed 1.

Name: MIN D-DIMENSIONAL VECTOR PACKING (MIN D-VP)

Instance: A set $A = \{a_1, \dots, a_n\}$ of n items, a number d of dimensions, and a number n of d -dimensional bins. Every item a_j has a d -dimensional positive rational size vector $s_j := (s_{1,j}, \dots, s_{d,j})$.

Task: Find n disjoint (possibly empty) subsets $A_1, \dots, A_n$ of A such that the number of non-empty subsets is minimized and the items of each subset fit into a bin.

For some instance I of MIN D-VP two items a_j and $a_{j'}$ can be put into the same subset (bin) if and only if

$$s_{1,j} + s_{1,j'} \leq 1 \wedge s_{2,j} + s_{2,j'} \leq 1 \wedge \dots \wedge s_{d,j} + s_{d,j'} \leq 1. \quad (4)$$

This motivates to model the compatibleness of the items in I by d -threshold graphs. For every dimension i , $1 \leq i \leq d$, we consider the inequality

$$s_{i,1}x_1 + s_{i,2}x_2 + \dots + s_{i,n}x_n \leq 1. \quad (5)$$

An instance I of MIN D-VP and a graph $G = (V, E)$ are *equivalent*, if there is a bijection $f : A \rightarrow V$ such that for every $A' \subseteq A$ the characteristic vector of A'

satisfies for every dimension i , $1 \leq i \leq d$, inequality (5) if and only if $f(A') := \{f(a_j) \mid a_j \in A'\}$ is an independent set of G .

If for every dimension i , $1 \leq i \leq d$, the MIN BP instance defined by inequality (5) has an equivalent graph $G_i = (V, E_i)$, see Sect. 3.3, we know that G_i is a threshold graph. Further, two items a_j and $a_{j'}$ can be put into the same bin if and only if

$$\{v_j, v_{j'}\} \notin E_1 \wedge \{v_j, v_{j'}\} \notin E_2 \wedge \cdots \wedge \{v_j, v_{j'}\} \notin E_d. \quad (6)$$

Then, $G = (V, E)$ with $E = E_1 \cup \cdots \cup E_d$ results in a d -threshold graph. By (6) two items a_j and $a_{j'}$ can be put into the same bin if and only if there is no edge $\{v_j, v_{j'}\}$ in G . That is, any two items corresponding to the vertices of a clique in G cannot be put into the same bin, i.e., $\omega(G) \leq OPT(I)$.

Since there is a partition of V into $\chi(G)$ non-empty independent sets V_i , the corresponding item sets $A_i = \{f^{-1}(v) \mid v \in V_i\}$ lead to a solution of I using $\chi(G)$ non-empty subsets A_i . Thus, we know $OPT(I) \leq \chi(G)$.

In order to obtain an equivalent graph for a MIN D-VP instance I , we can proceed as in Sect. 3.2.1 and use the graph $G_d(I)$ defined by the values of the d inequalities in (5).

Theorem 16 *Let I be an instance for MIN D-VP such that for every dimension i instance I_i has an equivalent graph $G_i = (V, E_i)$. Then, the graph $G = (V, E_1 \cup \cdots \cup E_d)$ leads to the bounds $\omega(G) = \omega(G_d(I)) \leq OPT(I) \leq \chi(G) = \chi(G_d(I))$.*

Next, we consider the generalization of MIN BP, where we have to pack items corresponding to d -dimensional parallelepipeds into bins corresponding to d -dimensional cubes of size 1 on every dimension. Each of the items a_j has for dimension i the size $s_{i,j}$. A set of items A' fits into a d -dimensional cube, if they can be placed without overlapping and without rotations into the cube, such that the faces of the items are parallel to those of the cube.

Name: MIN D-DIMENSIONAL BIN PACKING (MIN D-BP)

Instance: A set $A = \{a_1, \dots, a_n\}$ of n items, a number d of dimensions, and a number n of d -dimensional bins. Every item a_j has for dimension i the positive rational size $s_{i,j}$.

Task: Find n disjoint (possibly empty) subsets $A_1, \dots, A_n$ of A such that the number of non-empty subsets is minimized and the items of each subset fit into a d -dimensional cube.

A MIN D-BP instance I can be handled very similar to instances of MIN D-VP. The only difference is that we have to exchange in (4) and (6) every $\wedge$ by $\vee$ and in the definition of the edge set E of G we have to replace $\cup$ by $\cap$, resulting in a d -threshold intersection graph G .

In order to obtain an equivalent graph for a MIN D-BP instance I we can proceed similar to Sect. 3.2.1 and use $G'_d(I) = (V(I), E(I))$ defined by $V(I) = \{v_j \mid a_j \in A\}$ and $E(I) = \{\{v_j, v_{j'}\} \mid s_{1,j} + s_{1,j'} > c_1 \wedge \cdots \wedge s_{d,j} + s_{d,j'} > c_d\}$ using the values of the d inequalities in (5).

Table 3 Summary of results for counting and enumerating problems related to k -threshold graphs with n vertices and m edges

	Threshold graph G		k -threshold graph G	
$\alpha(G)$	$\mathcal{O}(n + m)$	Corollary 2	$\mathcal{O}\left(\sum_{i=1}^k m_i + k \cdot n \cdot \left(\prod_{i=1}^k \omega(G_i)\right)\right) \subseteq \mathcal{O}(k \cdot n^{k+1})$	Corollary 7
$\omega(G)$	$\mathcal{O}(n + m)$	By $\alpha(\overline{G})$	Open	
$\text{mis}(G)$	$\mathcal{O}(n + m)$	Corollary 3	$\mathcal{O}\left(\sum_{i=1}^k m_i + n \cdot \left(\prod_{i=1}^k \omega(G_i)\right)^2\right) \subseteq \mathcal{O}(n^{2k+1})$	Theorem 8
$\text{im}(G)$	$\mathcal{O}(n + m)$	Corollary 4	$\mathcal{O}\left(\sum_{i=1}^k m_i + k \cdot n \cdot \left(\prod_{i=1}^k \omega(G_i)\right)\right) \subseteq \mathcal{O}(k \cdot n^{k+1})$	Corollary 7
$\text{i}(G)$	$\mathcal{O}(n + m)$	Corollary 5	$\mathcal{O}(n^2 \cdot 2^n)$	Remark 1
$\text{mc}(G)$	$\mathcal{O}(n + m)$	By $\text{mis}(\overline{G})$	Open	
$\text{cm}(G)$	$\mathcal{O}(n + m)$	By $\text{im}(\overline{G})$	Open	
$\text{c}(G)$	$\mathcal{O}(n + m)$	By $\text{i}(\overline{G})$	Open	
$\text{MIS}(G)$	$\mathcal{O}(1)$	Corollary 3	$\mathcal{O}\left(\sum_{i=1}^k m_i + n \cdot \left(\prod_{i=1}^k \omega(G_i)\right)^2\right) \subseteq \mathcal{O}(n^{2k+1})$	Theorem 8
$\text{IM}(G)$	$\mathcal{O}(1)$	Corollary 4	$\mathcal{O}\left(\sum_{i=1}^k m_i + k \cdot n \cdot \left(\prod_{i=1}^k \omega(G_i)\right)\right) \subseteq \mathcal{O}(k \cdot n^{k+1})$	Corollary 7
$\text{I}(G)$	$\mathcal{O}(1)$	Corollary 5	$\mathcal{O}(n^2 \cdot 2^n)$	Remark 1
$\text{MC}(G)$	$\mathcal{O}(1)$	By $\text{MIS}(\overline{G})$	Open	
$\text{CM}(G)$	$\mathcal{O}(1)$	By $\text{IM}(\overline{G})$	Open	
$\text{C}(G)$	$\mathcal{O}(1)$	By $\text{I}(\overline{G})$	Open	

Running times for enumeration algorithms for threshold graphs are given in time per output

Theorem 17 *Let I be an instance of MIN D-BP such that for every dimension i instance I_i has an equivalent graph $G_i = (V, E_i)$. Then, the graph $G = (V, E_1 \cap \dots \cap E_d)$ provides the bounds $\omega(G) = \omega(G'_d(I)) \leq \text{OPT}(I) \leq \chi(G) = \chi(G'_d(I))$ for an optimal solution of I .*

4 Conclusions

We introduced methods to count and enumerate all maximal independent, maximum independent, and independent sets in threshold and k -threshold graphs. Our results for enumerating maximal independent sets in threshold graphs, split graphs, and 2-threshold graphs are better than existing solutions for general graphs (Tsukiyama et al. 1977). We were able to exploit these results in order to solve a large number of knapsack instances in polynomial time. Since we generated all maximal independent sets for our solutions, we can even extend the results of Theorems 11 and 13 to generate *all* optimal solutions of the respective knapsack instances. Further, since our solutions for knapsack problems using equivalent graphs are widely independent from the profits, they can be applied to the product knapsack problem (PKP) (D'Ambrosio et al. 2018).

Closely related problems concern the family of all cliques (maximum cliques and maximal cliques, respectively) of a graph G denoted by $\text{C}(G)$ ($\text{CM}(G)$ and $\text{MC}(G)$),

respectively). The cardinalities of these families are denoted by $c(G)$ ($cm(G)$ and $mc(G)$, respectively). Counting and listing maximal cliques in a threshold graph G can be done by considering maximal independent sets in the complement graph $\bar{G}$ via $MC(G) = MIS(\bar{G})$. The method described in the proof of Theorem 4 can be modified to enumerate all maximal cliques in a threshold graph.

In Table 3 we survey our results on counting and enumerating special independent sets and cliques within threshold graphs and k -threshold graphs.

The structure of k -threshold graphs turned out to be very useful for listing maximal independent sets. In a very similar way the structure of k -threshold intersection graphs, i.e., graphs whose threshold intersection dimension is at most k , can be used to list maximal cliques.

It remains an open question, how to list maximal cliques in k -threshold graphs and maximal independent sets in k -threshold intersection graphs. It would also be interesting to generalize the result of Theorem 9 to k -threshold graphs or even to (k, ℓ) -graphs, see Sect. 2.4, in order to improve the running time for listing maximal independent sets in k -threshold graphs, shown in Theorem 8. Furthermore, it remains open how to find good characterizations for the structure of (multidimensional) knapsack instances which have equivalent graphs. One of the main open questions is, whether one can verify in polynomial time if a given instance of the (multidimensional) knapsack problem has an equivalent graph.

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